Merge vs rebase

Objectives

- Describe how a merge joins diverged histories with a merge commit, while a fast-forward merge only moves the branch pointer
- Describe how a rebase replays your commits as new commits on a new base
- Apply the rule: do not rebase commits other people already have

Teaching

Start with a guess

While you worked on `feature`, `main` gained a commit. There are two ways to bring that work into your branch. The model shows the starting history.

Two ways to take in main

Both commands bring `main`'s new work into your branch.

Merge: `git merge main` keeps every existing commit and adds a merge commit with two parents.

Rebase: `git rebase main` replays your commits on top of `main`'s tip. Each replayed commit is new, with a new ID (C3 becomes C3′); the old ones are abandoned. History becomes a straight line.

The rule: Because rebase replaces commits, never rebase commits other people already have. Merging is always safe.

Apply it to a shared branch

Same two tools; now a teammate depends on your commits.

Model

Practice repository Simulation: no real Git runs

1. C4 Fix tax main
2. C3 Add search HEAD → feature
3. C2 Add cart

Recap

- `git merge main` keeps all commits and adds a merge commit with two parents.
- `git rebase main` replays your commits as new commits on main's tip, making a straight line.
- Rebased commits get new IDs; the old ones are abandoned.
- Never rebase commits others already have. When in doubt, merge.

Next topic: Pull requests and review comments

Cheat sheet

`git merge main`	join main into your branch with a merge commit
`git rebase main`	replay your commits on top of main
`C3 vs C3′`	same change, different commit after a rebase

Sources

Teaching claims were checked on 2026-10-06 (UTC) against live official documentation: the Git reference manual on git-scm.com (latest source release 2.56.0, released 2026-09-28). Simulated command output is illustrative: its exact wording was not checked and varies with Git version and settings.

- git-rebase: description
- git-rebase: recovering from upstream rebase
- git-merge: description

Static interaction alternatives

predict

Prediction (not scored). Your feature has C3; main gained C4 since you branched. On feature you run git rebase main. What happens to C3? Options: A new commit C3′ with the same change lands on top of C4; C3 itself moves, unchanged, on top of C4; C3 and C4 combine into one merge commit; C3 is deleted and its change lost. Answer: A new commit C3′ with the same change lands on top of C4. A new commit C3′ with the same change lands on top of C4: Rebase replays your change as a brand-new commit on main's tip. The old C3 is abandoned. C3 itself moves, unchanged, on top of C4: Attractive, but a commit's parent is part of its identity, so Git must create a new commit, C3′. C3 and C4 combine into one merge

commit: That describes a merge, not a rebase. C3 is deleted and its change lost: The change survives in C3′; only the old commit is left behind.

terminal

Practice terminal (simulated; nothing runs on your computer). Task: Try both from the same starting history and compare the results. Goals: Merge main into feature; find the merge commit; From the same start, rebase; find C3′. One sequence that meets every goal: git merge main, then Restore the starting history (practice only), then git rebase main.

mcq

Check (one retry, then the answer is shown). You pushed feature, and a teammate already built commits on top of it. You need main's new commits in feature. Which do you use? Options: git merge main; git rebase main; Either; both give the same history; git reset --hard main. Answer: git merge main. git merge main: Right: merge keeps every commit your teammate built on, adding one merge commit. git rebase main: Rebase replaces your pushed commits with new ones, so your teammate's work sits on commits that are no longer in your branch. Either; both give the same history: They differ: merge adds a merge commit; rebase creates new copies of your commits. git reset --hard main: That would throw away your feature commits entirely. Retry question: The model shows a history after someone took in main. Which command produced it? Options: git rebase main: C3 was replayed as C3′; git merge main: there is a merge commit; git merge main: C3 was kept as it was; Neither; nothing changed. Answer: git rebase main: C3 was replayed as C3′. git rebase main: C3 was replayed as C3′: Right: C3′ is a new commit on top of C4, and the old C3 is left with no branch. git merge main: there is a merge commit: No commit here has two parents, so no merge happened. git merge main: C3 was kept as it was: The branch now points at C3′, a different commit from C3. Neither; nothing changed: feature moved from C3 to C3′, so something did change.